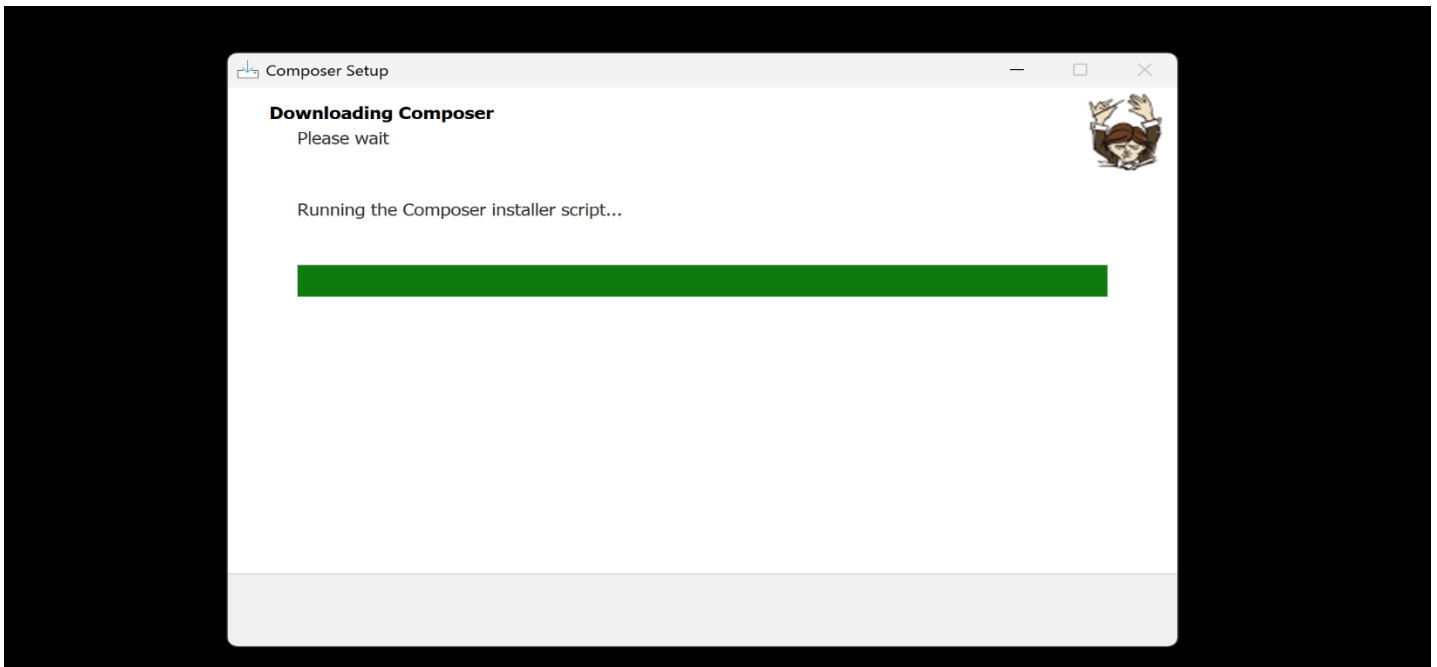


STEP 1 — INSTALL COMPOSER, DOWNLOAD CODEIGNITER 4

[Composer Official Website](https://getcomposer.org/)

Install Composer.

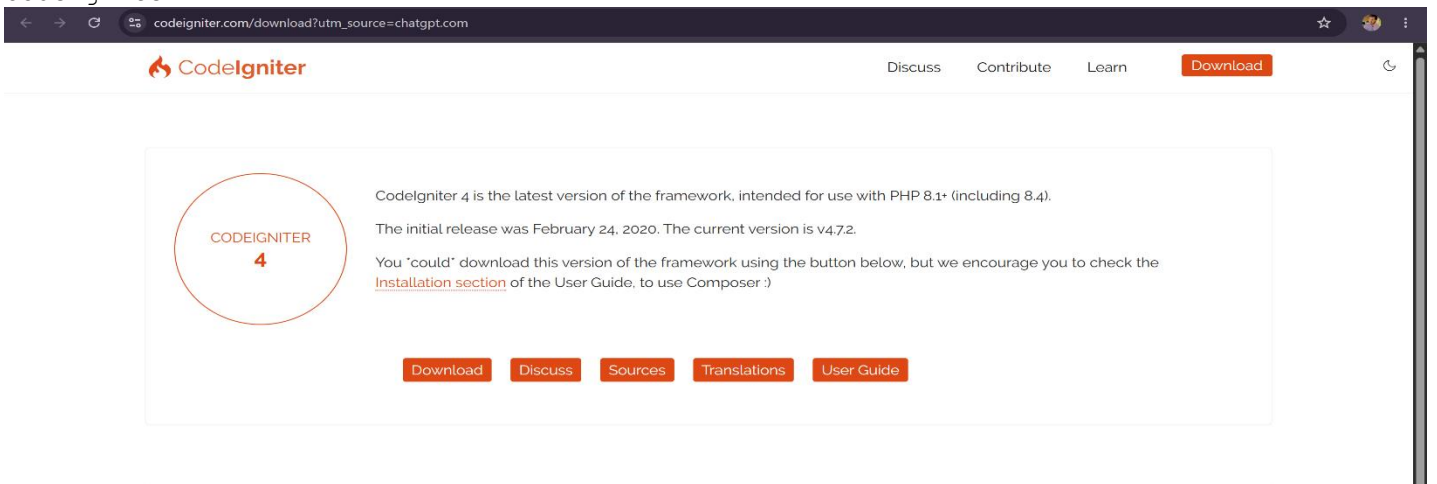


DOWNLOAD CODEIGNITER 4

[CodeIgniter Downloads](https://codeigniter.com/download/)

Download:

CodeIgniter 4



STEP 2 — CREATE PROJECT USING COMPOSER

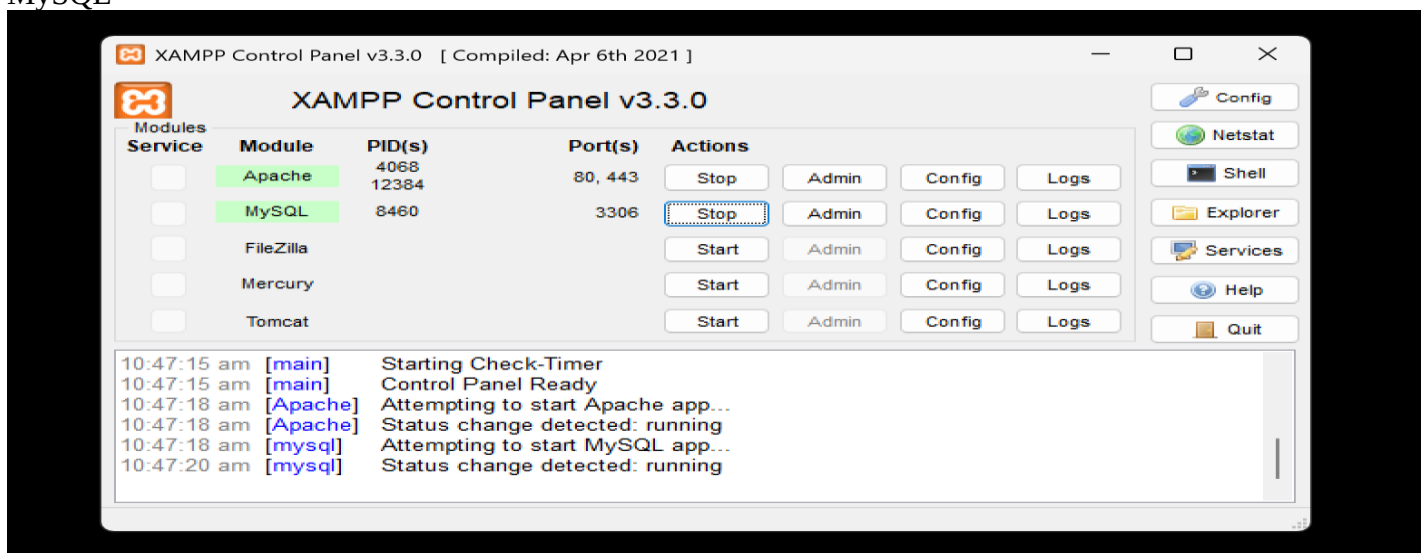
Open Command Prompt.

- `cd C:\xampp\htdocs`
- `composer create-project codeigniter4/appstarter lab12`

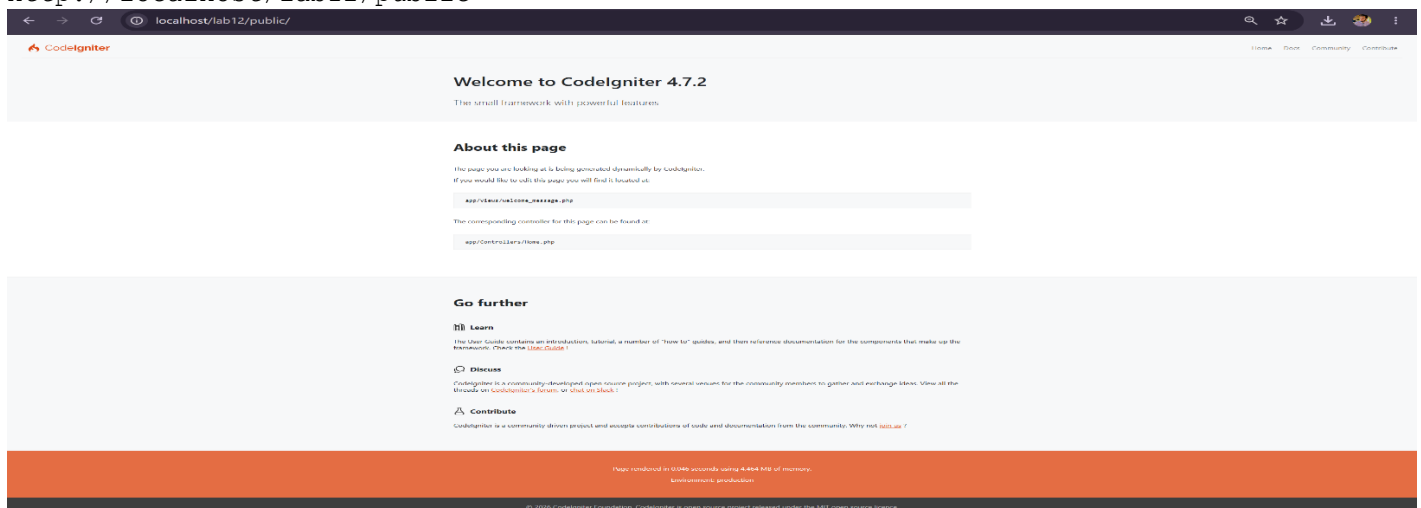
STEP 3 — START XAMPP

Start:

Apache
MySQL

**STEP 4 — TEST CODEIGNITER (In browser)**

`http://localhost/lab12/public`



STEP 5 — OPEN PROJECT IN VS CODE

Open:

C:\xampp\htdocs\lab12

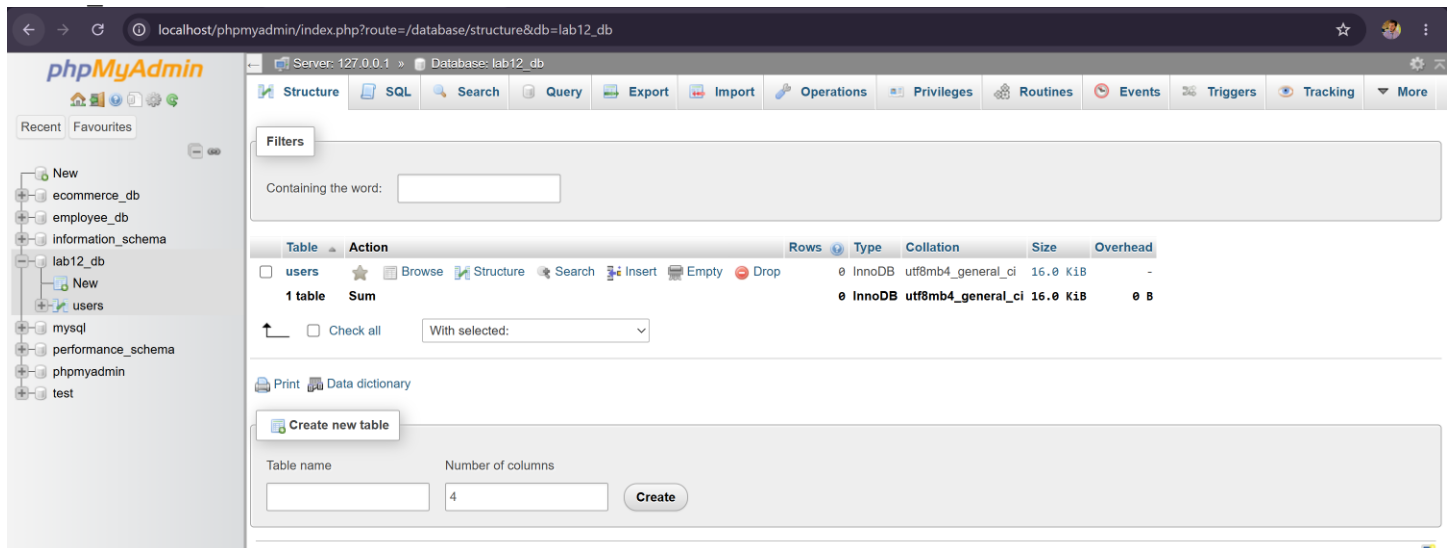
STEP 6 — CREATE DATABASE

Open browser:

http://localhost/phpmyadmin

Create database:

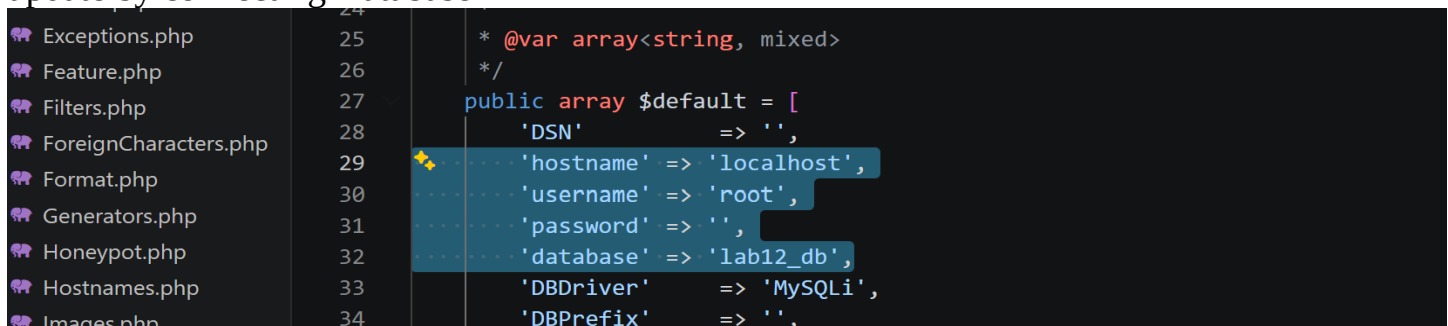
lab12 db

**STEP 8 — CONNECT DATABASE**

Open file:

app/Config/Database.php

update by connecting Database



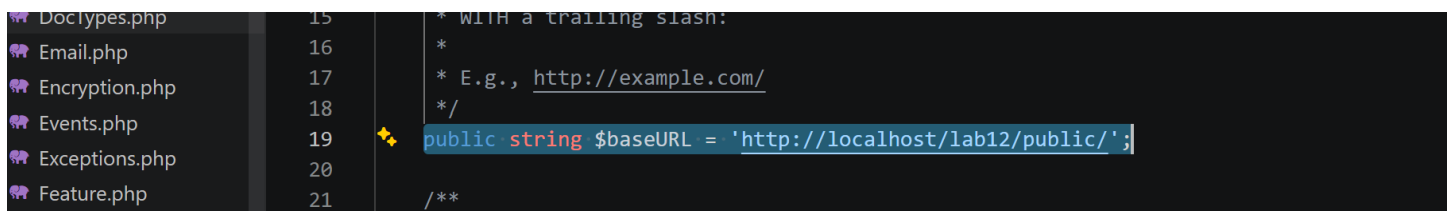
STEP 9 — SET BASE URL

app/Config/App.php

```
public string $baseUrl = 'http://localhost:8080/';
```

Replace with:

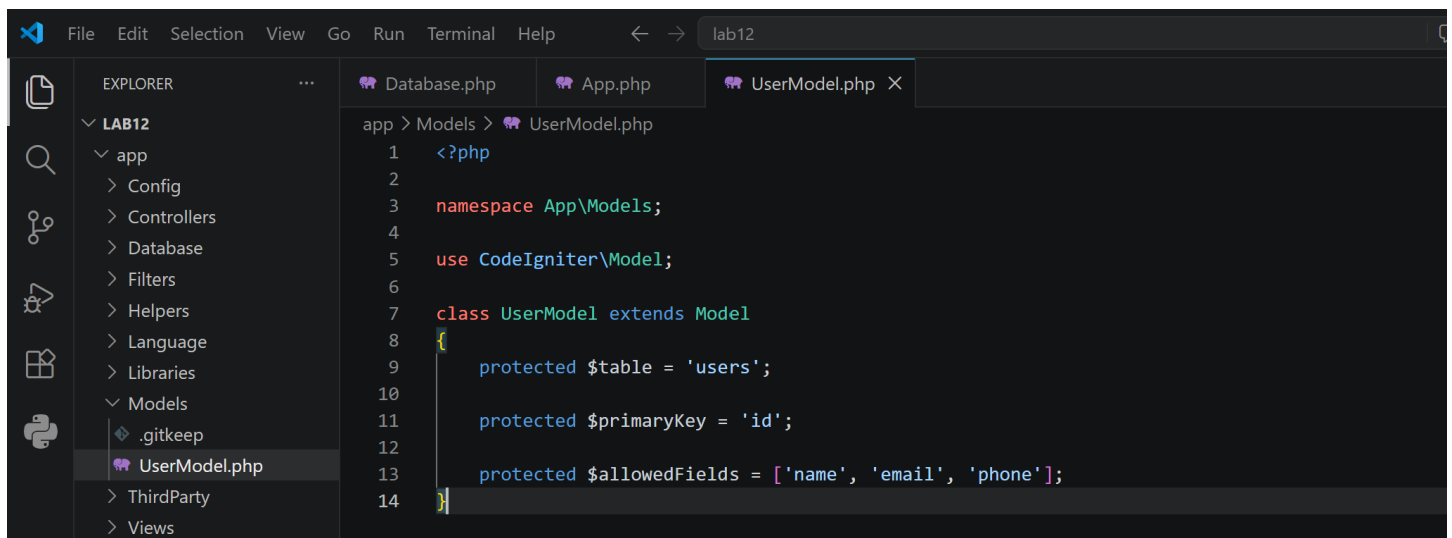
```
public string $baseUrl = 'http://localhost/lab12/public/';
```



STEP 10 — CREATE MODEL

Create file:

app/Models/UserModel.php



STEP 11 — CREATE CONTROLLER

Create file:

app/Controllers/UserController.php

```
LAB12
└─ app
   └─ Config
      └─ Controllers
         └─ BaseController.php
            └─ HomeController.php
               └─ UserController.php
                  └─ Database
                     └─ Filters
                        └─ Helpers
                           └─ Language
                              └─ Libraries
                                 └─ Models
                                    └─ ThirdParty
                                       └─ Views
                                          └─ errors
                                             └─ users
                                                └─ create.php
                                                   └─ edit.php
                                                      └─ index.php
                                                         └─ welcome_message.php
                                                            └─ .htaccess
                                                               └─ Common.php
                                                                  └─ index.html
                                                                     └─ public
                                                                        └─ tests
                                                                           └─ vendor
                                                                              └─ writable
                                                                                 └─ .gitignore
                                                                                    └─ builds
                                                                                       └─ composer.json
                                                                                          └─ composer.lock
                                                                                             └─ env
                                                                                                └─ LICENSE
                                                                                                   └─ phpunit.xml.dist
                                                                                                      └─ preload.php
                                                                                                         └─ README.md
                                                                                                            └─ OUTLINE
                                                                                                               └─ TIMELINE

app > Controllers > UserController.php
1  <?php
2
3  namespace App\Controllers;
4
5  use App\Models\UserModel;
6
7  class UserController extends BaseController
8  {
9      public function index()
10     {
11         $model = new UserModel();
12
13         $data['users'] = $model->findAll();
14
15         return view('users/index', $data);
16     }
17
18     public function create()
19     {
20         return view('users/create');
21     }
22
23     public function store()
24     {
25         helper(['form']);
26
27         $rules = [
28             'name' => 'required|min_length[3]',
29             'email' => 'required|valid_email',
30             'phone' => 'required|numeric|exact_length[11]'
31         ];
32
33         if (!$this->validate($rules)) {
34             return view('users/create', [
35                 'validation' => $this->validator
36             ]);
37         }
38
39         $model = new UserModel();
40
41         $model->save([
42             'name' => $this->request->getPost('name'),
43             'email' => $this->request->getPost('email'),
44             'phone' => $this->request->getPost('phone')
45         ]);
46
47         return redirect()->to('/users');
48     }
49
50     public function edit($id)
51     {
52         $model = new UserModel();
53
54         $data['user'] = $model->find($id);
55
56         return view('users/edit', $data);
57     }
58
59     public function update($id)
60     {
61         helper(['form']);
62
63         $rules = [
64             'name' => 'required|min_length[3]',
65             'email' => 'required|valid_email',
66             'phone' => 'required|numeric|exact_length[11]'
67         ];
68
69         if (!$this->validate($rules)) {
70             $model = new UserModel();
71
72             $data['user'] = $model->find($id);
73
74             $data['validation'] = $this->validator;
75
76             return view('users/edit', $data);
77         }
78     }
79 }
```

```
> Language
> Libraries
> Models
> ThirdParty
> Views
   └─ errors
      └─ users
         └─ create.php
            └─ edit.php
               └─ index.php
                  └─ welcome_message.php
                     └─ .htaccess
                        └─ Common.php
                           └─ index.html
                              └─ public
                                 └─ tests
                                    └─ vendor
                                       └─ writable
                                          └─ .gitignore
                                             └─ builds
                                                └─ composer.json
                                                   └─ composer.lock
                                                      └─ env
                                                         └─ LICENSE
                                                            └─ phpunit.xml.dist
                                                               └─ preload.php
                                                                  └─ README.md
                                                                     └─ OUTLINE
                                                                        └─ TIMELINE

45     ];
46
47     return redirect()->to('/users');
48 }
49
50 public function edit($id)
51 {
52     $model = new UserModel();
53
54     $data['user'] = $model->find($id);
55
56     return view('users/edit', $data);
57 }
58
59 public function update($id)
60 {
61     helper(['form']);
62
63     $rules = [
64         'name' => 'required|min_length[3]',
65         'email' => 'required|valid_email',
66         'phone' => 'required|numeric|exact_length[11]'
67     ];
68
69     if (!$this->validate($rules)) {
70         $model = new UserModel();
71
72         $data['user'] = $model->find($id);
73
74         $data['validation'] = $this->validator;
75
76         return view('users/edit', $data);
77     }
78 }
```

```
78 }
79
80 $model = new UserModel();
81
82 $model->update($id, [
83     'name' => $this->request->getPost('name'),
84     'email' => $this->request->getPost('email'),
85     'phone' => $this->request->getPost('phone')
86 ]);
87
88 return redirect()->to('/users');
89
90
91 public function delete($id)
92 {
93     $model = new UserModel();
94
95     $model->delete($id);
96
97     return redirect()->to('/users');
98 }
99 }
```

STEP 12 — CREATE users FOLDER

Create folder:

app/Views/users

STEP 13 — CREATE index.php

Create:

app/Views/users/index.php

```
EXPLORER    ...    Database.php    App.php    UserModel.php    UserController.php    index.php X
LAB12
├── app
│   ├── Config
│   ├── Controllers
│   │   ├── BaseController.php
│   │   ├── Home.php
│   │   └── UserController.php
│   ├── Database
│   ├── Filters
│   ├── Helpers
│   ├── Language
│   └── Libraries
└── app > Views > users > index.php
    1  <!DOCTYPE html>
    2  <html>
    3  <head>
    4  <title>User Management</title>
    5
    6  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">
    7  </head>
    8
    9  <body>
   10
   11  <div class="container mt-5">
   12
   13  <h2>User Management System</h2>
```

```
14
15 <a href="/users/create" class="btn btn-success mb-3">
16 Add User
17 </a>
18
19 <table class="table table-bordered table-striped">
20
21 <tr>
22 <th>ID</th>
23 <th>Name</th>
24 <th>Email</th>
25 <th>Phone</th>
26 <th>Actions</th>
27 </tr>
28
29 <?php foreach($users as $user): ?>
30
31 <tr>
32 |
33 <td><?= $user['id']; ?></td>
34 <td><?= $user['name']; ?></td>
35 <td><?= $user['email']; ?></td>
36 <td><?= $user['phone']; ?></td>
37
38 <td>
39
40 <a href="/users/edit/<?= $user['id']; ?>"
41 class="btn btn-primary btn-sm">
42 Edit
43 </a>
44
45 <a href="/users/delete/<?= $user['id']; ?>"
46 class="btn btn-danger btn-sm"
47 onclick="return confirm('Are you sure?')">
48 Delete
49 </a>
50
51 </td>
52
53 </tr>
54
```

```
55 </tr>
56
57 </table>
58
59 </div>
60
61 </body>
62 </html>
```

STEP 14 — CREATE create.php

Create:

app/Views/users/create.php

```
LAB12
├── app
│   ├── Config
│   ├── Controllers
│   │   ├── BaseController.php
│   │   ├── Home.php
│   │   └── UserController.php
│   ├── Database
│   ├── Filters
│   ├── Helpers
│   ├── Language
│   ├── Libraries
│   ├── Models
│   │   ├── .gitkeep
│   │   ├── UserModel.php
│   │   └── ThirdParty
│   ├── Views
│   │   ├── errors
│   │   ├── users
│   │   │   ├── create.php
│   │   │   ├── index.php
│   │   │   └── welcome_message.php
│   │   ├── .htaccess
│   │   ├── Common.php
│   │   ├── index.html
│   │   ├── public
│   │   ├── tests
│   │   └── vendor
├── Controllers
│   ├── BaseController.php
│   ├── Home.php
│   └── UserController.php
├── Database
├── Filters
├── Helpers
├── Language
├── Libraries
├── Models
│   ├── .gitkeep
│   ├── UserModel.php
│   └── ThirdParty
├── Views
│   ├── errors
│   ├── users
│   │   ├── create.php
│   │   ├── index.php
│   │   └── welcome_message.php
│   ├── .htaccess
│   ├── Common.php
│   ├── index.html
│   ├── public
│   ├── tests
│   └── vendor
├── writable
├── .gitignore
├── builds
├── composer.json
└── ...

app > Views > users > create.php
1  <!DOCTYPE html>
2  <html>
3  <head>
4  <title>Add User</title>
5
6  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">
7  </head>
8
9  <body>
10
11  <div class="container mt-5">
12
13  <h2>Add User</h2>
14
15  <form method="post" action="/users/store">
16
17  <div class="mb-3">
18  <label>Name</label>
19
20  <input type="text" name="name" class="form-control">
21
22  <?php if(isset($validation)): ?>
23  <small class="text-danger">
24  <?=$validation->getError('name'); ?>
25  </small>
26  <?php endif; ?>
27
28  </div>
29
30  <div class="mb-3">
31  <label>Email</label>
32
33  <input type="text" name="email" class="form-control">
34
35  <?php if(isset($validation)): ?>
36  <small class="text-danger">
37  <?=$validation->getError('email'); ?>
38  </small>
39  <?php endif; ?>
40
41  </div>
42
43  <div class="mb-3">
44  <label>Phone</label>
45
46  <input type="text" name="phone" class="form-control">
47
48  <?php if(isset($validation)): ?>
49  <small class="text-danger">
50  <?=$validation->getError('phone'); ?>
51  </small>
52  <?php endif; ?>
53
54  </div>
55
56  <button class="btn btn-success">
57  Save User
58  </button>
59
60  </form>
61
62  </div>
63
64  </body>
65  </html>
```


STEP 15 — CREATE edit.php

Create:

app/Views/users/edit.php

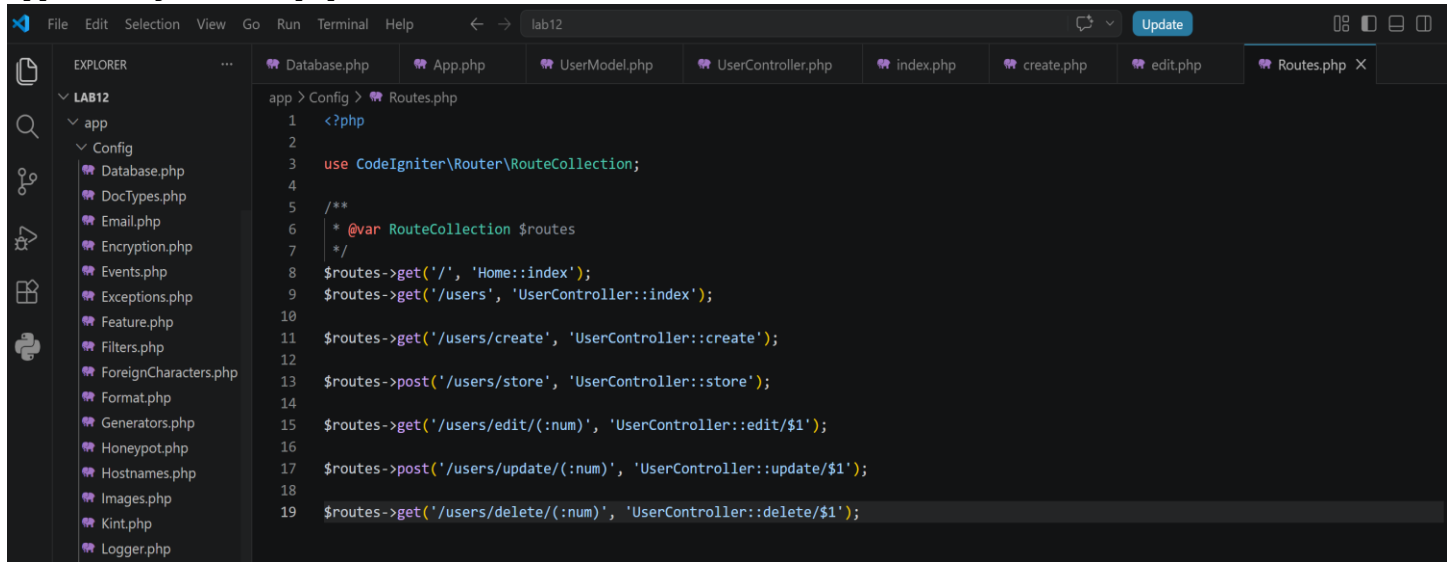
The screenshot shows a code editor with a file explorer on the left and a code editor on the right. The file explorer shows a project structure with folders like 'app', 'controllers', 'views', and 'models'. The code editor displays the 'edit.php' file, which contains PHP code for editing a user. The code includes HTML form elements for name, email, and phone, and PHP logic for validation and database updates.

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4 <title>Edit User</title>
5
6 <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">
7 </head>
8
9 <body>
10
11 <div class="container mt-5">
12
13 <h2>Edit User</h2>
14
15 <form method="post" action="/users/update/<? $user['id']; ?>">
16
17 <div class="mb-3">
18 <label>Name</label>
19
20 <input type="text"
21 name="name"
22 class="form-control"
23 value="<? $user['name']; ?>">
24
25 <?php if(isset($validation)): ?>
26 <small class="text-danger">
27 <? $validation->getError('name'); ?>
28 </small>
29 <?php endif; ?>
30
31 </div>
32
33 <div class="mb-3">
34 <label>Email</label>
35
36 <input type="text"
37 name="email"
38 class="form-control"
39 value="<? $user['email']; ?>">
40
41 <?php if(isset($validation)): ?>
42 <small class="text-danger">
43 <? $validation->getError('email'); ?>
44 </small>
45 <?php endif; ?>
46
47 </div>
48
49 <div class="mb-3">
50 <label>Phone</label>
51
52 <input type="text"
53 name="phone"
54 class="form-control"
55 value="<? $user['phone']; ?>">
56
57 <?php if(isset($validation)): ?>
58 <small class="text-danger">
59 <? $validation->getError('phone'); ?>
60 </small>
61 <?php endif; ?>
62
63 </div>
64
65 <button class="btn btn-primary">
66 Update User
67 </button>
68
69 </form>
70
71 </div>
72
73 </body>
```

STEP 16 — ADD ROUTES

Open:

app/Config/Routes.php



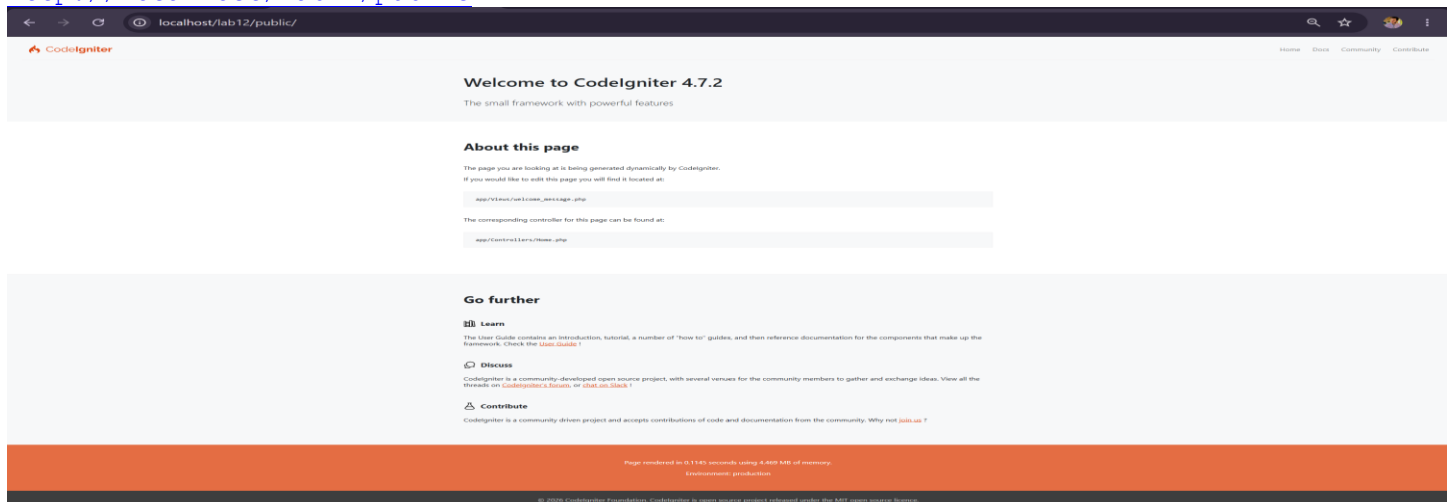
```
1 <?php
2
3 use CodeIgniter\Router\RouteCollection;
4
5 /**
6  * @var RouteCollection $routes
7  */
8 $routes->get('/', 'Home::index');
9 $routes->get('/users', 'UserController::index');
10
11 $routes->get('/users/create', 'UserController::create');
12
13 $routes->post('/users/store', 'UserController::store');
14
15 $routes->get('/users/edit/(:num)', 'UserController::edit/$1');
16
17 $routes->post('/users/update/(:num)', 'UserController::update/$1');
18
19 $routes->get('/users/delete/(:num)', 'UserController::delete/$1');
```

STEP 17 — OUTPUT

In browser

1. CodeIgniter Welcome Page

<http://localhost/lab12/public>



http://localhost/phpmyadmin

The top screenshot shows the phpMyAdmin interface for the 'lab12_db' database. The 'Structure' tab is selected, showing the 'users' table. The table has one column named 'id' of type 'InnoDB utf8mb4_general_ci' with a size of 16.0 KiB. The bottom screenshot shows the 'Table: users' view with one row of data: id=1, name='Rifaq Ajmal', email='rifaqajmal@gmail.com', and phone='03432128540'.

3. Empty User Table

http://localhost/lab12/public/users

The screenshot shows the 'User Management System' web application. It features a green 'Add User' button and a table with columns: ID, Name, Email, Phone, and Actions.

ID	Name	Email	Phone	Actions
----	------	-------	-------	---------

4. Validation Errors Screenshot

A browser window showing the 'Add User' form at localhost/lab12/public/users/create. The form has three input fields: 'Name' with the value 'osoanwioq', 'Email' with the value 'gewgrehre', and 'Phone' with the value '0896634643'. The 'Phone' field is highlighted with a blue border. A green 'Save User' button is at the bottom.

A browser window showing the 'Add User' form at localhost/lab12/public/users/store. The form has three input fields: 'Name', 'Email', and 'Phone'. The 'Email' field has a red error message below it: 'The email field must contain a valid email address.' The 'Phone' field has a red error message below it: 'The phone field must be exactly 11 characters in length.' A green 'Save User' button is at the bottom.

5. Add User Form

A browser window showing the 'Add User' form at localhost/lab12/public/users/create. The form has three empty input fields: 'Name', 'Email', and 'Phone'. A green 'Save User' button is at the bottom.

localhost/lab12/public/users/create

Add User

Name
Rifaq Ajmal

Email
rifaqajmalmohmand@gmail.com

Phone
03432128540

Save User

6. User Added Successfully

localhost/lab12/public/users

User Management System

Add User

ID	Name	Email	Phone	Actions
1	Rifaq Ajmal	rifaqajmalmohmand@gmail.com	03432128540	Edit Delete

7. Edit User Page

Edit User

Name
Talha

Email
rifaqajmalmohmand@gmail.com

Phone
03432128540

Update User

8. Updated Record

User Management System

Add User

ID	Name	Email	Phone	Actions
1	Talha	rifaqajmalmohmand@gmail.com	03432128540	Edit Delete



User Management System

Add User

ID	Name	Email	Phone	Actions
----	------	-------	-------	---------

10. Final Table



User Management System

Add User

ID	Name	Email	Phone	Actions
2	Talha	rifaqajmalmohmand@gmail.com	03432128540	Edit Delete